

# Toric Ideals of Non-Bipartite Graphs on Eight Vertices

*When is the edge ring of a graph failing the odd cycle condition Cohen–Macaulay or Gorenstein?*

Ibrahim Rashid and Navila Zaman

Adviser: Professor Augustine O’Keefe · Department of Mathematics · Connecticut College · Summer Science Research Institute

## The toric ideal of a graph

Let  $G$  be a finite simple graph with vertices  $v_1, \dots, v_n$  and edges  $e_1, \dots, e_m$ . Over a field  $k$ , define the ring map

$$\varphi : k[e_1, \dots, e_m] \rightarrow k[v_1, \dots, v_n], \quad e = \{v_i, v_j\} \mapsto v_i v_j.$$

The **toric ideal** of  $G$  is  $I_G = \ker \varphi$ , and the **edge ring** is  $k[G] = k[e_1, \dots, e_m]/I_G$ . Bipartite graphs are well understood, so we study **connected non-bipartite** graphs; for these,  $\dim k[G] = n$ .

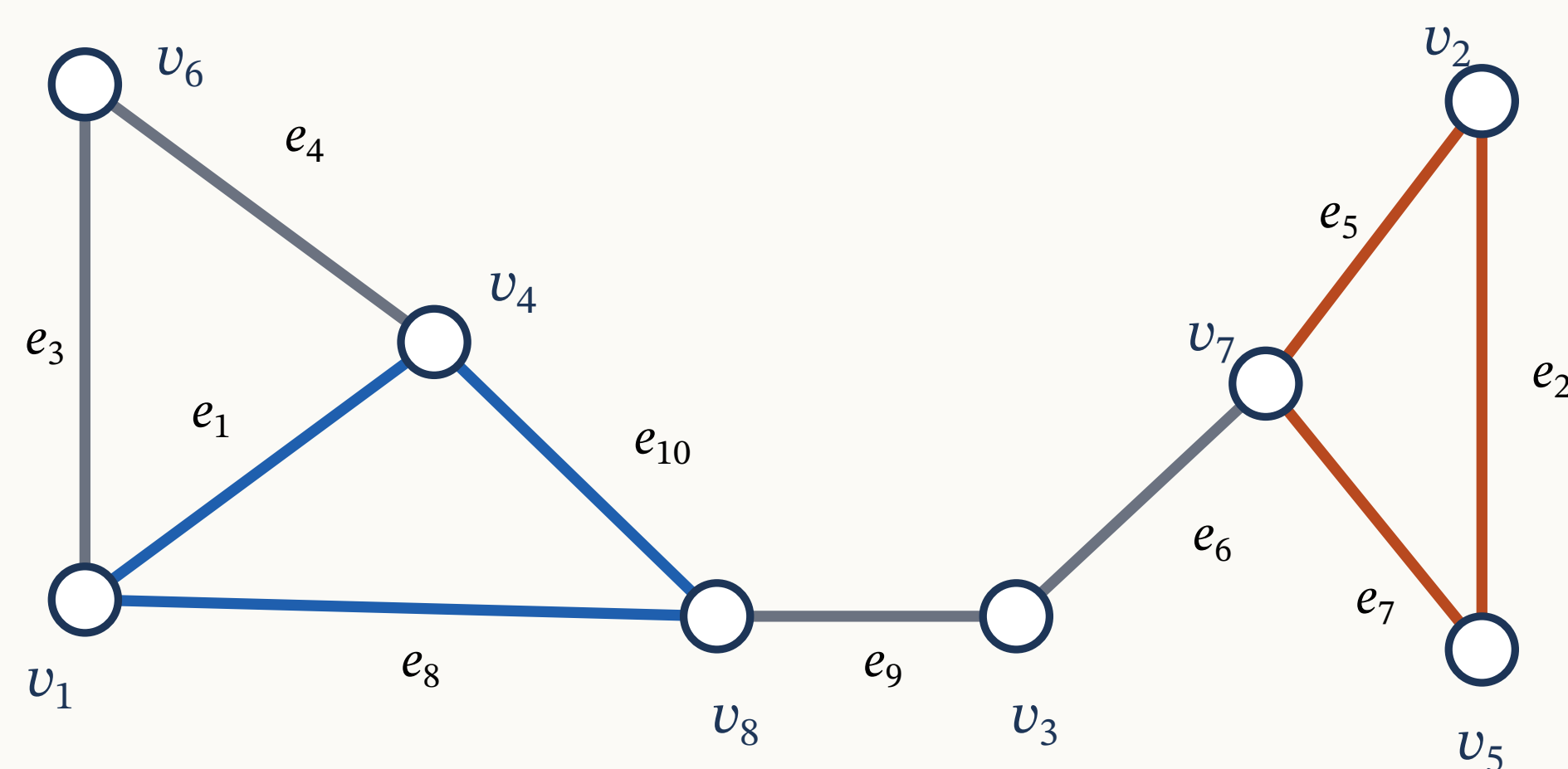
**Theorem 1** (Villarreal 1995; Ohsugi–Hibi 1999)

The ideal  $I_G$  is generated by the binomials

$$f_w = e_{i_1} e_{i_3} \cdots e_{i_{2s-1}} - e_{i_2} e_{i_4} \cdots e_{i_{2s}},$$

one for each even closed walk  $w = (e_{i_1}, \dots, e_{i_{2s}})$  in  $G$ .

## Example: two generators



Graph #2305 ( $n = 8$ ,  $m = 10$ ). The triangles  $\mathbf{C} = v_1 v_4 v_8$  and  $\mathbf{C}' = v_2 v_5 v_7$  share no vertex and no edge joins them.

By Theorem 1,  $I_G = (g_1, g_2)$  with

$$g_1 = e_4 e_8 - e_3 e_{10}, \quad g_2 = e_1 e_5 e_7 e_9^2 - e_2 e_6^2 e_8 e_{10}.$$

$g_1$  comes from the 4-cycle  $v_1 v_8 v_4 v_6$ ;  $g_2$  from the closed walk of length 10 around  $C$ , along  $v_8 v_3 v_7$ , around  $C'$  and back. Since  $2 = m - n$ ,  $I_G$  is a complete intersection, and

$$H(k[G], t) = \frac{1 + 2t + 2t^2 + 2t^3 + 2t^4 + t^5}{(1 - t)^8}.$$

The numerator is symmetric, so  $k[G]$  is Gorenstein, although  $G$  fails the odd cycle condition.

### Definition (Odd cycle condition)

A graph  $G$  satisfies the **odd cycle condition (OCC)** if for any two vertex-disjoint odd cycles  $C$  and  $C'$  of  $G$ , some edge of  $G$  joins a vertex of  $C$  to a vertex of  $C'$ . A pair with no joining edge is a **separated pair**;  $G$  fails OCC exactly when it has one.

**Theorem 2** (Ohsugi–Hibi 1998; Simis–Vasconcelos–Villarreal 1998)

For a connected graph  $G$ , the edge ring  $k[G]$  is normal if and only if  $G$  satisfies the odd cycle condition.

**Theorem 3** (Hochster 1972)

Every normal affine semigroup ring is Cohen–Macaulay.

**Consequence.** If  $G$  satisfies OCC, then  $k[G]$  is Cohen–Macaulay. The converse is **false**: the graph in the Example fails OCC, yet  $k[G]$  is Cohen–Macaulay. The open question is which graphs **failing** OCC have Cohen–Macaulay edge rings.

### Definitions (Cohen–Macaulay, Hilbert series)

Let  $R = k[e_1, \dots, e_m]$ . The ring  $R/I_G$  is **Cohen–Macaulay** if  $\text{depth}(R/I_G) = \dim(R/I_G) = n$ . By the Auslander–Buchsbaum formula,  $\text{depth} = m - \text{pd}(R/I_G)$ , so depth can be read off a minimal free resolution.

The Hilbert series has the reduced form

$$H(R/I_G, t) = \frac{h_0 + h_1 t + \cdots + h_s t^s}{(1 - t)^n},$$

with  $h$ -vector  $(h_0, \dots, h_s)$ ,  $h_0 = 1$  and  $h_1 = m - n$ . If  $R/I_G$  is Cohen–Macaulay, then every  $h_i \geq 0$ .

**Theorem 4** (Stanley 1978)

A Cohen–Macaulay standard graded domain is Gorenstein if and only if its  $h$ -vector is symmetric:  $h_i = h_{s-i}$  for all  $i$ .

Since  $k[G]$  is a domain, a Cohen–Macaulay edge ring with symmetric  $h$ -vector is Gorenstein. Every complete intersection is Gorenstein.

### Theorem 5 (Edge threshold; N. Zaman)

Let  $G$  be a connected graph with  $n \geq 7$  vertices and  $m$  edges. If  $G$  fails the odd cycle condition, then

$$n + 1 \leq m \leq \frac{n(n-1)}{2} - 9,$$

and every value of  $m$  in this range occurs. If  $n \leq 6$ , every connected graph satisfies OCC.

## Proof

**Upper bound.** Let  $C, C'$  be a separated pair with  $a$  and  $b$  vertices. None of the  $ab$  pairs with one vertex in each cycle is an edge, and  $a, b \geq 3$ , so  $G$  misses at least  $ab \geq 9$  of the  $n(n-1)/2$  possible edges.

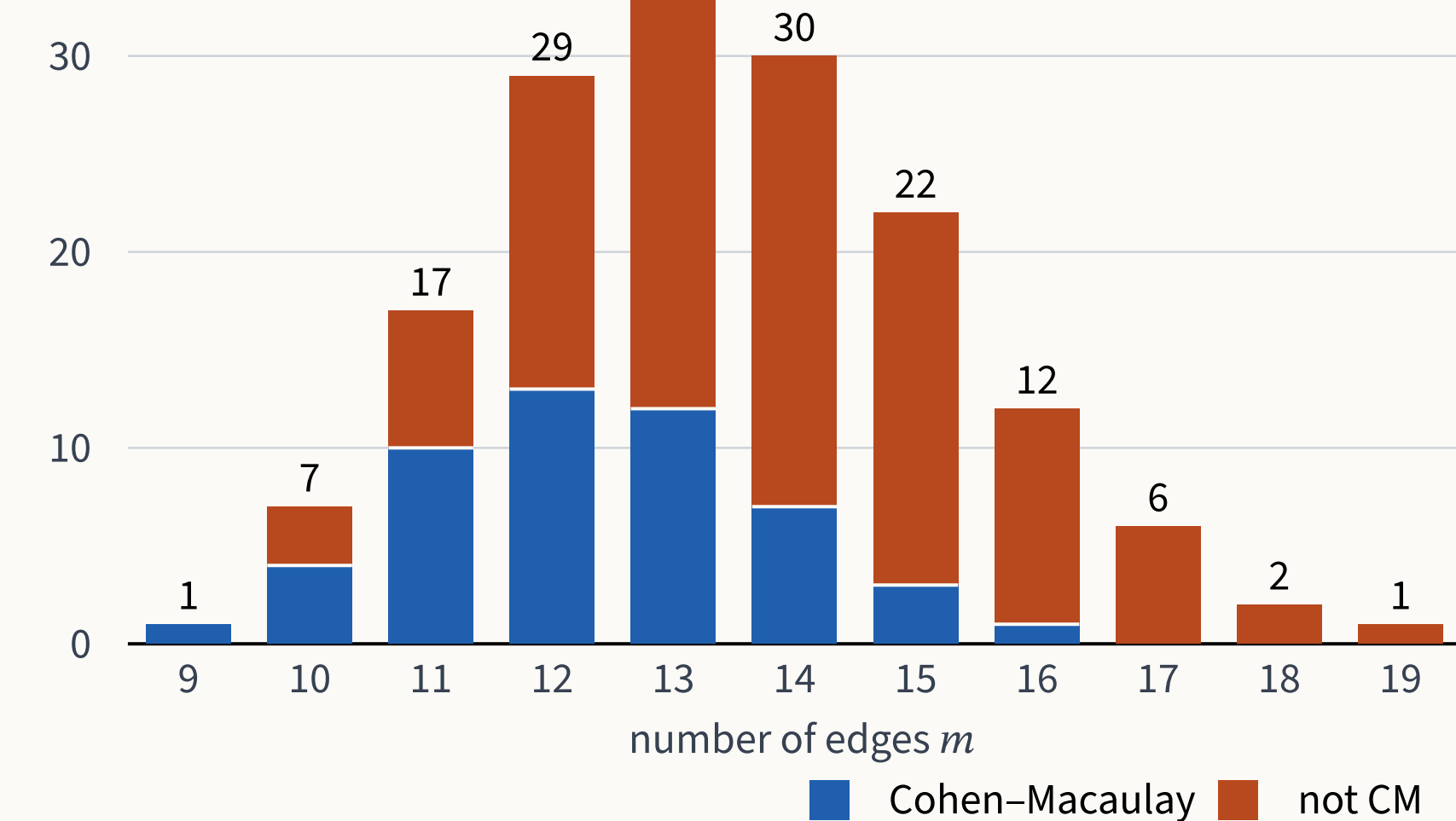
**Lower bound.** Contract  $C$  and  $C'$  to single vertices. The result is connected with  $n - a - b + 2$  vertices, so  $G$  has at least  $n - a - b + 1$  edges outside  $C \cup C'$ . Adding the  $a + b$  cycle edges gives  $m \geq n + 1$ .

**Every value occurs.** Two triangles joined by a path through the other  $n - 6$  vertices has  $n + 1$  edges and fails OCC. Add, one at a time, every edge that does not join the two triangles: each graph is connected and fails OCC, ending at  $n(n-1)/2 - 9$  edges.

**Small  $n$ .** Two vertex-disjoint odd cycles need at least 6 vertices; when  $n = 6$  they cover every vertex, so connectivity forces an edge between them. ■

## What it gives for $n = 8$

A graph on 8 vertices can fail OCC only if  $9 \leq m \leq 19$ . Every graph with 20 or more edges is Cohen–Macaulay by Theorems 2 and 3 and needs no computation. The bound is attained at both ends:



The 161 graphs on 8 vertices (connected, minimum degree  $\geq 2$ ) that fail OCC, by number of edges.

## Computation

We listed every connected graph on 8 vertices up to isomorphism (nauty, through the NautyGraphs package of Macaulay2) and kept the non-bipartite graphs of minimum degree  $\geq 2$ ; a leaf only adds a free variable to  $k[G]$ . By Theorem 5 only  $9 \leq m \leq 19$  matters: **6,971** graphs. For each, Macaulay2 computed  $I_G = \ker \varphi$ ,  $\text{depth}(R/I_G)$  and the Hilbert series. The odd cycle condition was checked separately for every graph.

## Results for $n = 8$

| Graphs with $9 \leq m \leq 19$    | count      |
|-----------------------------------|------------|
| satisfy OCC (CM by Theorems 2, 3) | 6,810      |
| fail OCC                          | 161        |
| ... Cohen–Macaulay                | <b>51</b>  |
| ... not Cohen–Macaulay            | <b>110</b> |
| Gorenstein (Theorem 4)            | 435        |

- Every non-Cohen–Macaulay edge ring has depth  $7 = n - 1$ , and its  $h$ -vector has a negative entry, e.g.  $(1, 2, 2, 2, 2, -1)$  for graph #89.
- 32% of the graphs failing OCC are Cohen–Macaulay; none with  $m \geq 17$  is.
- If a separated pair lies in one 2-connected block of  $G$ , then  $k[G]$  is not Cohen–Macaulay (92 of 92 graphs), consistent with Hà–Kara–O’Keefe [7, Thm 5.1].
- Of the 435 Gorenstein edge rings, 432 come from graphs satisfying OCC. Exactly three graphs fail OCC, and all three are complete intersections:

| graph | $m$ | generator degrees | $h$ -vector          |
|-------|-----|-------------------|----------------------|
| #1979 | 9   | 6                 | $(1, 1, 1, 1, 1, 1)$ |
| #2305 | 10  | 2, 5              | $(1, 2, 2, 2, 2, 1)$ |
| #5222 | 11  | 2, 2, 4           | $(1, 3, 4, 4, 3, 1)$ |

## Future work

- Extend the classification to 9 vertices (7,125 graphs fail OCC) and 10 vertices (596,940 graphs fail OCC); these computations are under way.
- Find the structure of separated odd cycles that decides whether  $k[G]$  is Cohen–Macaulay.
- Determine when a Gorenstein edge ring of a graph failing OCC is a complete intersection.

## Acknowledgements

We thank Professor Augustine O’Keefe for guidance throughout this project, and the Connecticut College Summer Science Research Institute for its support.

## References

- [1] R. H. Villarreal, Rees algebras of edge ideals, *Comm. Algebra* 23 (1995).
- [2] H. Ohsugi and T. Hibi, Toric ideals generated by quadratic binomials, *J. Algebra* 218 (1999).
- [3] H. Ohsugi and T. Hibi, Normal polytopes arising from finite graphs, *J. Algebra* 207 (1998).
- [4] A. Simis, W. V. Vasconcelos and R. H. Villarreal, The integral closure of subrings associated to graphs, *J. Algebra* 199 (1998).

- [5] M. Hochster, Rings of invariants of tori, Cohen–Macaulay rings generated by monomials, and polytopes, *Ann. of Math.* 96 (1972).
- [6] R. P. Stanley, Hilbert functions of graded algebras, *Adv. Math.* 28 (1978).
- [7] H. T. Hà, S. Kara and A. O’Keefe, Algebraic properties of toric rings of graphs, arXiv:1703.08270 (2017).
- [8] D. R. Grayson and M. E. Stillman, Macaulay2; B. D. McKay and A. Piperno, nauty.